SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

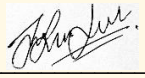
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	**Total Marks:**	100
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
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Section / Batch:	Slot A	**Date:**	23-10-26

Code of Conduct

I Joel Benn John B(1911260003) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Mobile IP Tunnel Overhead

A logistics company uses **Mobile IP** to allow delivery vehicles to remain connected while moving between cities.
The communication between the **home agent** and **foreign agent** is established using a tunneling mechanism.
The tunneling process adds an overhead of **20 bytes per packet** during transmission.
One vehicle sends a **500-byte data packet** to the headquarters.
Calculate the additional overhead introduced by the tunnel for this packet.

ANSWER:

Given

- Original data packet size = 500 bytes
- Tunnel overhead = 20 bytes per packet

Formula

Additional overhead = overhead added per packet

Calculation

The tunnel adds 20 bytes to the original packet.
Total packet size = 500 + 20 = 520 bytes.
Overhead percentage = (20 / 500) × 100 = 4%.
Final answer: Additional overhead = 20 bytes; total packet size = 520 bytes; overhead = 4%.
Key concept: Mobile IP tunneling encapsulates a packet inside another packet, adding header bytes.

2. Maximum Theoretical Data Rate

A university campus deploys a wireless network to support high-speed multimedia applications.
The network operates with a **bandwidth of 20 MHz** and uses **64-QAM modulation**.
Each transmitted symbol carries **6 bits** of information.
The administrator wants to estimate the network's maximum theoretical throughput.
Calculate the maximum theoretical data rate of the wireless network.

ANSWER :

Given

- Bandwidth = 20 MHz
- Modulation = 64-QAM
- Bits per symbol = 6

Formula

Simplified estimate: Data rate = bandwidth × bits per symbol

Calculation

Convert bandwidth: 20 MHz = 20×10^6 Hz.
Data rate = $20 \times 10^6 \times 6 = 120 \times 10^6$ bits/s.
Therefore, the simplified theoretical estimate is 120 Mbps.
Final answer: 120 Mbps (simplified theoretical estimate).

Key concept: 64-QAM has 64 possible symbol states, so each symbol carries $\log_2(64) = 6$ bits. Actual throughput depends on symbol rate, coding, and protocol overhead.

3. Routing Table Memory Requirement

A proactive routing protocol is implemented in a military communication network to ensure routes are always available.
Each mobile node maintains a routing table containing **50 entries**.
Every routing entry occupies **32 bytes** of memory.
The network administrator needs to estimate the memory consumed by the routing table.
Calculate the total memory required for storing the routing table.

ANSWER :

Given

- Number of routing entries = 50
- Memory per entry = 32 bytes

Formula

Total memory = number of entries × memory per entry

Calculation

Total memory = 50 × 32 = 1600 bytes.
Using 1 KiB = 1024 bytes: 1600 / 1024 = 1.5625 KiB.
Final answer: 1600 bytes = 1.5625 KiB.
Key concept: A proactive routing protocol maintains routes in advance, and each node uses memory to store its routing table.

4. Distance Travelled Between Routing Updates

An emergency response vehicle communicates through an **ad-hoc wireless network** while moving across a disaster zone.
The vehicle travels at a constant speed of **10 m/s**.
The routing protocol updates its routing information every **2 seconds**.
The system designer wants to know how far the vehicle moves before each update occurs.
Calculate the total distance traveled between consecutive routing updates.

ANSWER :

Given

- Vehicle speed = 10 m/s
- Update interval = 2 seconds

Formula

Distance = speed × time

Calculation

Distance = 10 × 2 = 20 metres.
Final answer: 20 metres.
Key concept: At a constant speed, distance travelled between updates is the speed multiplied by the time interval.

5. DSR Route Discovery

A rescue operation uses the **Dynamic Source Routing (DSR)** protocol to establish communication between two mobile devices.
During route discovery, the route request packet passes through **5 intermediate nodes** before reaching the destination.
Each node forwards the request only once to continue the discovery process.
The network engineer wants to determine the number of route request packets involved in this process.
Calculate the total number of route request packets generated.

ANSWER:

Given

- Intermediate nodes = 5
- Each intermediate node forwards the request once

Formula

Total RREQ transmissions = source transmission + forwarded transmissions

Calculation

The source sends one Route Request (RREQ).
Each of the five intermediate nodes forwards it once: 5 transmissions.
Total = 1 + 5 = 6 RREQ transmissions.
Final answer: 6 RREQ transmissions, under the stated assumption.
Key concept: DSR is an on-demand routing protocol. The count assumes one source transmission and one forwarding transmission per intermediate node; real networks may suppress duplicates.

6. Mobile IP Subnet Addressing

A Mobile IP-based enterprise network is divided into multiple subnets for efficient address management.
One subnet is configured with the subnet mask **255.255.255.240**.
The network administrator plans to allocate IP addresses to mobile nodes within this subnet.
Before deployment, the maximum number of usable host addresses must be determined.
Calculate how many mobile nodes can be assigned IP addresses in one subnet.

ANSWER:

Given

- Subnet mask = 255.255.255.240

Formula

Usable IPv4 hosts = 2^(number of host bits) − 2

Calculation

The subnet mask has 28 network bits, so the prefix is /28.
Host bits = 32 − 28 = 4.
Total addresses = $2^4 = 16$.
Usable host addresses = 16 − 2 = 14 (excluding network and broadcast addresses).
Final answer: 14 usable host addresses.

Key concept: A conventional IPv4 /28 subnet contains 16 total addresses, usually with 14 available for hosts.

7. Spectral Efficiency

An IoT-enabled smart factory operates its wireless communication system at **2.4 GHz**.
The communication channel has a **bandwidth of 20 MHz**.
The transmitter operates at a **symbol rate of 1 Msps**.
The engineer needs to evaluate how efficiently the available bandwidth is utilized.
Calculate the spectral efficiency of the wireless communication system.

ANSWER :

Given

Carrier frequency = 2.4 GHz

Bandwidth = 20 MHz

Symbol rate = 1 Msps = 1×10^6 symbols/s

Formula

Spectral efficiency = data rate / bandwidth (bits/s/Hz)

Calculation

The symbol-rate-to-bandwidth ratio is $(1 \times 10^6) / (20 \times 10^6) = 0.05$ symbols/s/Hz.

This is not, by itself, the bit spectral efficiency. The modulation scheme or data rate is needed to determine bits/s/Hz.

For example, if each symbol carries 1 bit, data rate = 1 Mbps and spectral efficiency = 1 Mbps / 20 MHz = 0.05 bit/s/Hz.

Final answer: 0.05 symbols/s/Hz as a symbol-rate ratio. Actual bit/s/Hz requires the modulation scheme or data rate.

Key concept: Carrier frequency does not directly determine spectral efficiency; spectral efficiency compares data rate with channel bandwidth.

8. Network Density

A wildlife monitoring project deploys an **ad-hoc wireless sensor network** across a protected forest area.
The network consists of **50 sensor nodes**, each having a **transmission range of 250 meters**.
The total deployment area is **1 km²**.
The project team wants to estimate the density of nodes within the network.
Calculate the approximate network density.

ANSWER :

Given

Number of sensor nodes = 50

Deployment area = 1 km²

Transmission range = 250 m

Formula

Node density = number of nodes / deployment area

Calculation

Node density = 50 / 1 = 50 nodes/km².

Optional coverage estimate: radius = 250 m = 0.25 km.

Circular coverage area per node = $\pi r^2 = \pi \times (0.25)^2 \approx 0.1963$ km².

Sum of 50 individual coverage areas before overlap = $50 \times 0.1963 \approx 9.82$ km². This is not unique covered area because coverage regions can overlap.

Final answer: 50 nodes/km².

Key concept: Node density is the number of deployed nodes per unit area. Transmission range is not needed for the basic density calculation.

9. Doppler Shift

A high-speed train provides onboard wireless connectivity while traveling through a cellular network.

The train moves at a speed of **30 m/s**.

The wireless communication system operates at a carrier frequency of **900 MHz**.

The network engineer needs to evaluate the effect of motion on the received signal frequency.

Calculate the Doppler shift experienced by the mobile node.

ANSWER :

Given

Vehicle speed = 30 m/s

Carrier frequency = 900 MHz

Speed of light $c \approx 3 \times 10^8$ m/s

Formula

Maximum Doppler shift magnitude: $f_d = (v / c) \times f_c$

Calculation

Convert carrier frequency: 900 MHz = 900×10^6 Hz.

$f_d = [30 / (3 \times 10^8)] \times (900 \times 10^6) = 90$ Hz.

The shift is positive when approaching and negative when moving directly away, under the maximum-shift assumption.

Final answer: Maximum Doppler shift magnitude = 90 Hz (approximately ±90 Hz depending on direction).

Key concept: Doppler shift changes the received frequency due to relative motion between transmitter and receiver.

10. Packet Transmission Time

A company uses a **wireless LAN** to transfer large engineering design files between workstations.

The WLAN supports a data rate of **54 Mbps**.
Each transmitted packet has a size of **1500 bytes**.
The network administrator wants to estimate the transmission delay for a single packet.
Calculate the transmission time required to transmit one packet.

ANSWER :

Given

- Data rate = 54 Mbps
- Packet size = 1500 bytes

Formula

Transmission time = packet size in bits / data rate in bits per second

Calculation

Convert packet size to bits: $1500 \times 8 = 12{,}000$ bits.
Convert data rate: 54 Mbps = 54×10^6 bits/s.
Time = $12{,}000 / (54 \times 10^6) = 0.0002222$ seconds.
Convert to microseconds: $0.0002222 \times 10^6 \approx 222.22$ μs.
Final answer: Approximately 222.22 microseconds.
Key concept: This is transmission time only; it excludes propagation delay, WLAN headers and preambles, contention, and retransmissions.